Language on the move: Migration and linguistic diversity

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Population movements are a key force in language evolution, as they shape opportunities for language contact. While migration is widely acknowledged as a driver of linguistic change, its relationship to linguistic diversity has rarely been quantified in terms of magnitude on a global scale. This study investigates how international migration contributes to linguistic diversity at the country level. Linguistic diversity is measured using linguistic entropy. Alongside total linguistic entropy, which reflects the internal linguistic composition of a country, this study intro-duces imported linguistic entropy, a measure of migration-driven linguistic input derived from bilateral migrant stock data (1990–2020). Linear regression models were used to estimate the strength and stability of the association between imported and total linguistic entropy. Results show a strong and stable positive relationship: countries receiving migrants from linguistically diverse origins tend to display higher total linguistic diversity. These findings provide global quantitative evidence on the extent to which migration contributes to linguistic diversification and plays a measurable role in contemporary language evolution.

1. Introduction

Population movements are a central factor in language evolution, as they create conditions for language contact (Mufwene, 2007). This is particularly relevant today in what has been described as the "Age of Migration" (Borlongan, 2023). When speakers of different languages come into contact through migration, languages influence each other in their use, structure, and distribution. While such contact can increase linguistic diversity in a specific area, it can also lead to a re-duction in diversity due to language shift, convergence, and ultimately language loss (Gavin et al., 2013), making migration a key driver of the decline of global linguistic diversity (Loh & Harmon, 2014). In evolutionary terms, migration and contact condition variation and selection (Mufwene, 2007; Lim, 2010).

While the role of migration in language evolution is widely acknowledged, most research has focused on contact-induced structural change, such as borrowing, convergence, or the emergence of new contact varieties (Mufwene & Vigouroux, 2017; Güldemann & Hammarström, 2020). However, little is known about how migration affects the distribution and relative presence of languages

within a given region. Studies that link migration to linguistic diversity mainly examine prehistoric or historical population movements and specific regions (Nettle, 1999; Nichols, 2008; Gavin & Sibanda, 2012), whereas the magnitude and temporal stability of the contemporary relationship between migration and linguistic diversity have not yet been quantified at a global scale.

This study provides a global quantitative analysis of migration and linguistic diversity at the country level. Linguistic diversity is operationalized using linguistic entropy, which accounts for both the number of languages and their relative proportions within a population. A central contribution of this study is the computation of imported linguistic entropy, a measure that captures the linguistic impact of migration by combining migration stock data with the linguistic diversity of origin countries. Based on this measure, linear regression models are used to estimate the strength and directionality of migration effects globally.

2. Global migration developments

International migration is defined as "the movement of persons away from their place of usual residence and across an international border"; a person qualifies as a migrant if they have stayed in their new place of residence for at least 12 months (Sironi, Bauloz, & Emmanuel, 2019, p. 113). In 2024, 304 million people were international migrants, representing 3.7% of the world's population, up from 154 million (2.9%) in 1990 (International Organization for Migration, 2024, p. 20). Although migrants constitute a relatively small share of the global population, migration has profound impacts on both sending and receiving communities (Borlongan, 2023).

2.1 Migration stock dataset

Migrant stocks refer to the number of people living outside their country of birth (or citizenship) at a given point in time, whereas migrant flows capture the number of people who changed their country of residence within a specific time period (Ha et al., 2025). This study uses international migrant stock data, as they more accurately reflect the composition of a country's population at a specific moment.

Table 1. Top 5 international migration corridors ranked by 2024 migrant stock shown for selected years.

Destination (ISO)	Origin (ISO)	1990	2000	2010	2024
United States (US)	Mexico (MX)	4,506,017	9,245,920	11,895,927	11,279,561
Iran (IR)	Afghanistan (AF)	3,980,307	2,232,283	2,572,558	3,752,317
Turkey (TR)	Syria (SY)	5,234	5,357	46,891	3,563,983
Ukraine (UA)	Russia (RU)	5,018,098	3,745,842	3,265,811	3,375,095
United Arab Emirates (AE)	India (IN)	456,794	888,393	2,168,857	3,248,545

The international migrant stock dataset provided by the United Nations Population Division covers the years 1990 to 2024 in five-year intervals (with 2024 as the most recent update) and includes 233 countries, disaggregated by sex. The primary data source is population and housing censuses, complemented by population registers, nationally representative surveys, and administrative records (International Organization for Migration, 2024). For the purpose of this study, the dataset was enriched with ISO 3166-1 alpha-2 country codes (International Organization for Standardization, 2013), and male and female counts were aggregated to obtain total international migrant stocks per country.

3. Linguistic entropy as a measure of linguistic diversity

Linguistic diversity can be described in different ways. Language richness captures the number of languages in a given area but ignores their relative sizes. Phylogenetic diversity considers genealogical relations between languages (Gavin et al., 2013), but is more suited to historical and typological questions and lies beyond the scope of this study. Shannon entropy, a measure derived from information theory and ecology, is used here as it accounts for both the number of languages and their proportional distribution within a population and is widely recommended for measuring linguistic diversity due to its mathematical properties (Grin & Fürst, 2022).

Linguistic entropy was computed using country-level L1 speaker data from Ethnologue (Eberhard, Simons, & Fennig, 2023) and the Joshua Project (Joshua Project, 2023). Due to the lack of diachronic speaker data, linguistic entropy could only be computed for 2023, resulting in a static measure. Speaker counts were normalized by the total number of speakers per country, which then served as input for calculating *total linguistic entropy* for 238 countries as

$$H = -\sum_{i=1}^{N} p_i \log p_i \tag{1}$$

where p_i is the proportion of speakers of language i, N the total number of languages in the country, and where $\log$ refers to the natural logarithm (Shannon, 1948).

Entropy is lowest when one language dominates the population, and it increases as speakers are more evenly distributed across multiple languages.

3.1 Imported linguistic entropy

Building on the concept of linguistic entropy, this study distinguishes between two dimensions of linguistic diversity: total linguistic entropy and imported

linguistic entropy. Total linguistic entropy reflects the internal linguistic composition of a country, while imported linguistic entropy captures the linguistic diversity introduced through migration. This measure serves as an approximation of the diversity potentially introduced through migration, rather than directly measuring the language repertoires of migrant populations.

Imported linguistic entropy is based on the bilateral migration stock data described in the previous section. For each destination country and year (1990-2020), the relative share of migrants from each origin country was calculated. These migration shares were then used as weights for the total linguistic entropy values of the origin countries, reflecting the linguistic diversity that migrant populations potentially introduce into the destination country, so that, for each destination country, *imported linguistic entropy* is computed as

$$I = \sum_{c \in C} H_c m_c \tag{2}$$

where C collects all origin countries, and where m_c denotes the migration share of origin country c in the destination country and H_c is total linguistic entropy of c as in equation (1). Due to the static nature of the speaker data, the total linguistic entropy values from 2023 were used to estimate imported linguistic entropy across all migration years.

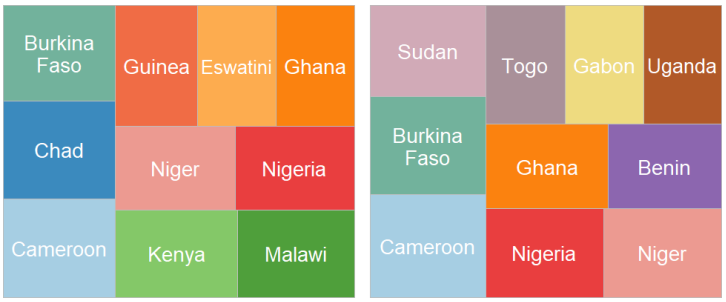

Figure 1. Treemaps of the top 12 countries by imported linguistic entropy in 1990 (left) and 2020 (right). Area size reflects the relative contribution of migration to linguistic diversity in each country.

Together, these measures make it possible to distinguish between internally maintained diversity and migration-driven diversity, linking present-day linguistic patterns to ongoing evolutionary dynamics shaped by population movement.

4. The impact of migration on linguistic entropy

To examine the association between imported and total linguistic diversity, seven cross-sectional linear regression models were estimated (Pearson, 1896), one for each reported migration year (1990–2020, 5-year intervals). Total linguistic

entropy in 2023 served as the dependent variable and imported linguistic entropy of the respective year as the predictor, using a consistent estimation procedure across years[1]. Although the migration dataset extends to 2024, only data up to 2020 were used, ensuring that the predictor does not postdate the outcome. Each model estimates a single cross-national association across countries, with countries as the units of observation, rather than country-specific effects.

Imported linguistic entropy is a strong and consistent predictor of total linguistic entropy across all years (Fig. 2). Estimated coefficients range from 0.84 (1990) to 0.93 (2015), and all effects are highly significant ($p < 0.001$; Table 2). Model fit, assessed using the coefficient of determination (R^2) and root mean squared error (RMSE) (Montgomery, Peck, & Vining, 2001), indicates moderate explanatory power: R^2 ranges from 0.30 to 0.37[2], meaning imported entropy ex-plains about one third of cross-national variation in linguistic diversity, while the remaining unexplained variance (RMSE ≈ 0.76) points to additional structural in-fluences. Overall, the stable effect sizes across time suggest a robust relationship: countries receiving migrants from linguistically diverse origins tend to exhibit systematically higher linguistic diversity.

Table 2. Imported entropy predicting total entropy. Across all years, imported entropy is positively associated with total entropy; model fit indicates moderate explanatory power.

Year	Estimate	*p*-value	R^2	RMSE
1990	0.846	< .001	0.34	0.77
1995	0.836	< .001	0.30	0.79
2000	0.889	< .001	0.33	0.78
2005	0.886	< .001	0.33	0.78
2010	0.920	< .001	0.35	0.77
2015	0.929	< .001	0.36	0.76
2020	0.910	< .001	0.37	0.76

To ensure the robustness of these findings, model assumptions and sensitivity to data structure were examined (Zuur, Ieno, & Elphick, 2010). Residual and Q–Q plots were inspected to confirm linearity and normality. In addition, two alternative model specifications were tested: a generalized linear model with a Gamma distribution and log link was fitted, and a model excluding outliers was estimated. In both cases, the direction, magnitude, and significance of the relationship between imported and total linguistic entropy remained stable, confirming that the results are not driven by distributional assumptions or extreme cases.

[1]While it may appear circular to include entropy values from 2023 in both the outcome and the predictor, circularity does not arise since imported linguistic entropy solely relies on the total linguistic entropy of other countries.

[2]R^2 increases slightly over time, likely because total entropy is derived from 2023 speaker data, making imported entropy a closer approximation of the underlying population structure in later years.

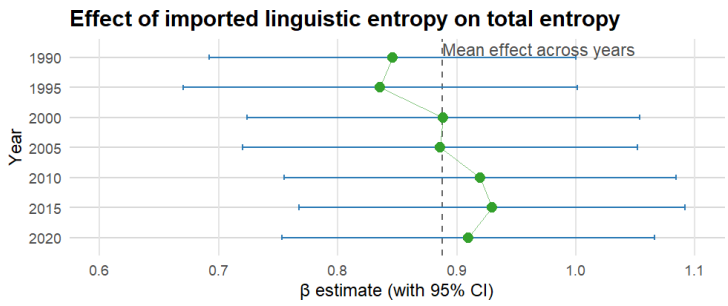

Figure 2. Yearly linear regression estimates (1990–2020) with 95% confidence intervals. The dashed line indicates the mean effect across years.

5. Conclusion and outlook

This study provides global quantitative evidence on the strength and temporal stability of the association between migration and contemporary patterns of linguistic diversity. Imported linguistic entropy is a strong and stable predictor of total linguistic entropy across all examined years (1990–2020), substantiating that countries receiving migrants from linguistically diverse origins indeed tend to maintain higher levels of overall linguistic diversity (Vertovec, 2007).

Several limitations must be acknowledged. Total entropy was treated as a static measure due to the lack of diachronic speaker data, which restricts the temporal interpretation of the results. The regression analysis identifies associations rather than causal effects; the results therefore do not imply that migration directly causes changes in linguistic diversity (Pearl, 2009). Indeed, it is not implausible that linguistic diversity also attracts migrants (Adserà & Pytliková, 2015), and that other factors not considered in this study, such as education, urbanization, or economic development, may simultaneously shape linguistic diversity. The analysis was conducted at the country level. While motivated by data availability and policy relevance, this approach inevitably abstracts from subnational variation and transnational dynamics (Anderson, 2019). Finally, migration stock data include reporting inconsistencies and gaps, particularly for undocumented migration, which affects measurement precision (Ha et al., 2025).

Future work should address these limitations by incorporating diachronic linguistic data once available, by extending the analysis with causal modeling approaches that account for confounding factors, and by moving beyond

national aggregates toward transnational and urban case studies. Despite its limitations, this study quantifies the extent to which migration is systematically associated with linguistic diversity and provides a scalable basis for investigating the demo-graphic drivers of language evolution.

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